

ebaleytherogue's No Man's Sky Bases Report Generator Script

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What is This?

This is a Python script that analyzes one's gamesave file's JSON export to generate a list of player bases. As a bonus I added the portal location for each base to the data, making it simple to share any base's location, that you wish, with other players. Thus, with this data you can:

- Know how many bases you have (and how many in each specific galaxy)
- Quickly share your base location with others without having to manually type the addresses into chat or journey to the base and take screenshots

To create the reports I heavily leaned on the JSON file's Teleporter entries. This got me what I needed. I attempted to give information about how many base objects exist, within any given base, but I could not get consistently accurate counts, because that information is scattered throughout the JSON, for any given base, and I did not want to provide inaccurate counts. I felt nothing is better than something in that regard.

Frankly, the hardest part of this script was calculating the bases' portal addresses hex values to compute properly. I'm not a math guy, but with AI's help I got it done. I used <https://nmscd.com/NMSGlyphRandomizer/?ref=nmscd> pasting the 12 digit hex addresses into that site's 'Enter glyphs' field to visually check my values against the portal glyphs, and I achieved a 100% success rate with what I saw in Camera Mode whilst on planet.

For you, this means you could copy and paste any value, from the reports' "Glyph String (No Spaces)" column, into that web site's field, and get the visual glyphs for that base.
Easy peasy.

Feel free to share this script, change this script, do whatever you like with this script. It was an exercise for me, kind of a side-quest in the game itself. Have fun with it!

I will keep it in the [ebaleytherogue's No Man's Sky_All Your Bases via Python Script_SHA...](#) shared folder unless I end up moving it to GitHub or something, although if I do, I'll drop a note in the folder with that location.

What this Isn't

This is not a gamesave editor. It does not manipulate your gamesave file in anyway. It simply interrogates a JSON export and produces report files.

Using the Report Files

The report files are simply csv (comma separated values) formatted text files that you could read in a text editor. However, Google Sheets, Excel, Libre Office, etc., know to read CSV files as spreadsheets. So, simply drag the .csv files to a folder in your Google Docs, and then open them. You can then view/sort/manipulate them as you would any spreadsheet.

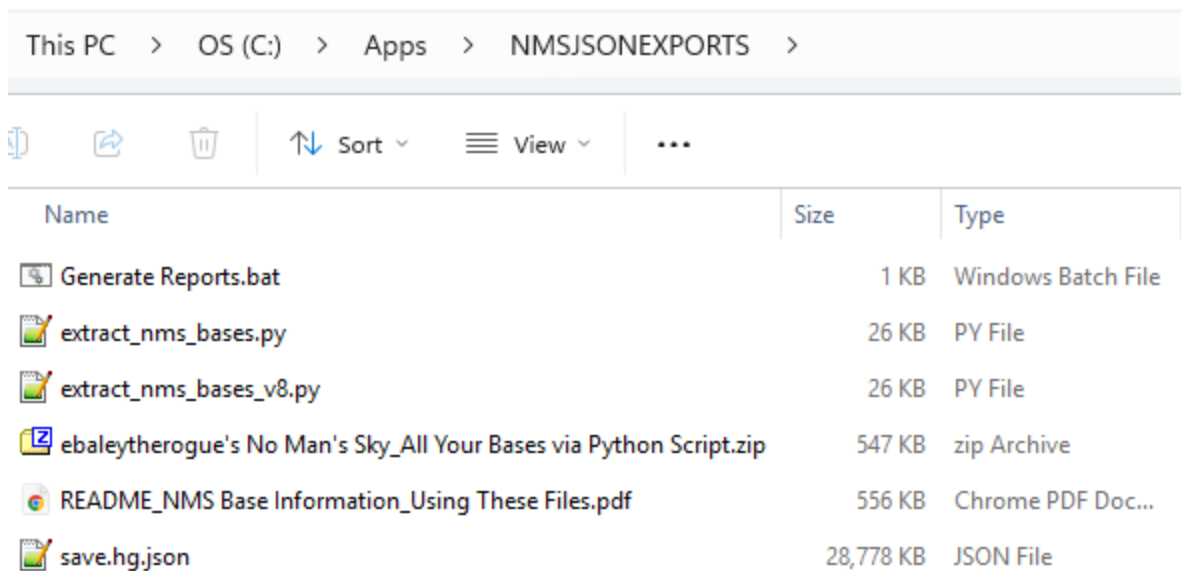
Using These Files to Create Reports

Prerequisites

In Windows 11 you will need Python capability and PowerShell capability to run it. I suppose you could just use a good old Windows command line window if you'd rather, but I used PowerShell. This document is not a PowerShell nor Python tutorial, so follow your googles and AI's and you'll get there. All you have to do is enable these things and/or perform a small installation to get them up and running. You do not need to program anything! At worst you'll need to right-click-open the .bat file to change the name of your JSON export.

Steps

Extract the files to a folder in your Windows system. I suggest something like shown below, but it can be anywhere.



The screenshot shows a Windows File Explorer window with the address bar set to 'This PC > OS (C:) > Apps > NMSJSONEXPORTS >'. The toolbar includes icons for navigation and actions, with 'Sort' and 'View' dropdown menus. The file list contains the following items:

Name	Size	Type
 Generate Reports.bat	1 KB	Windows Batch File
 extract_nms_bases.py	26 KB	PY File
 extract_nms_bases_v8.py	26 KB	PY File
 ebaleytherogue's No Man's Sky_All Your Bases via Python Script.zip	547 KB	zip Archive
 README_NMS Base Information_Using These Files.pdf	556 KB	Chrome PDF Doc...
 save.hg.json	28,778 KB	JSON File

There will likely be two Python script files (.py) in the ZIP file you download, as shown above. They are identical. One does not contain the version number due to reasons outlined in changelog item [Changed Main Script Name to NOT Include Version Number and .BAT file to Reference This New Name](#).

The example is using a gamesave filename of save.hg.json, so rename your actual JSON file to match this prior to running the script. You can generate a JSON by using Goatfungus' app or the newer Vectorcmdr app, both named 'NMSSaveEditor'. You can probably use NomNom

too, although I haven't tested that one.

I used a March 2026 build of Vectorcmdr and an older Goatfungus version to test and vet this script.

Once the files are in your folder of choice, simply double-click the .bat file to produce your reports. Example:

```
C:\WINDOWS\system32\cmd. X + v
C:\Apps\NMSJSONEXPORTS>python .\extract_nms_bases.py .\save.hg.json
C:\Apps\NMSJSONEXPORTS\extract_nms_bases.py:314: SyntaxWarning: "\"T" is an invalid escape sequence. Such sequences will
not work in the future. Did you mean "\"T"? A raw string is also an option.
"C:\Temp\Stuff" <-- invalid JSON, but some exporters still emit it
Standard JSON parse failed.
Reason: Invalid \escape: line 402772 column 15 (char 6952292)
Attempting automatic repair of invalid backslashes inside strings...
Master CSV written: C:\Apps\NMSJSONEXPORTS\nms_base_reports\nms_bases_master.csv
Grouped CSV written: C:\Apps\NMSJSONEXPORTS\nms_base_reports\nms_bases_grouped.csv
Duplicate report written: C:\Apps\NMSJSONEXPORTS\nms_base_reports\nms_bases_duplicate_names.csv
Summary written: C:\Apps\NMSJSONEXPORTS\nms_base_reports\nms_bases_summary.txt

Total unique teleporter-base rows: 203
Repeated base-name groups: 2

C:\Apps\NMSJSONEXPORTS>pause
Press any key to continue . . .
```

Ignore the syntax warning. It is the result of the script examining the JSON file and not totally digging it, but the warnings are simply warnings. The output is created successfully into a subfolder called nms_base_reports:

This PC > OS (C:) > Apps > NMSJSONEXPORTS > nms_base_reports_v3

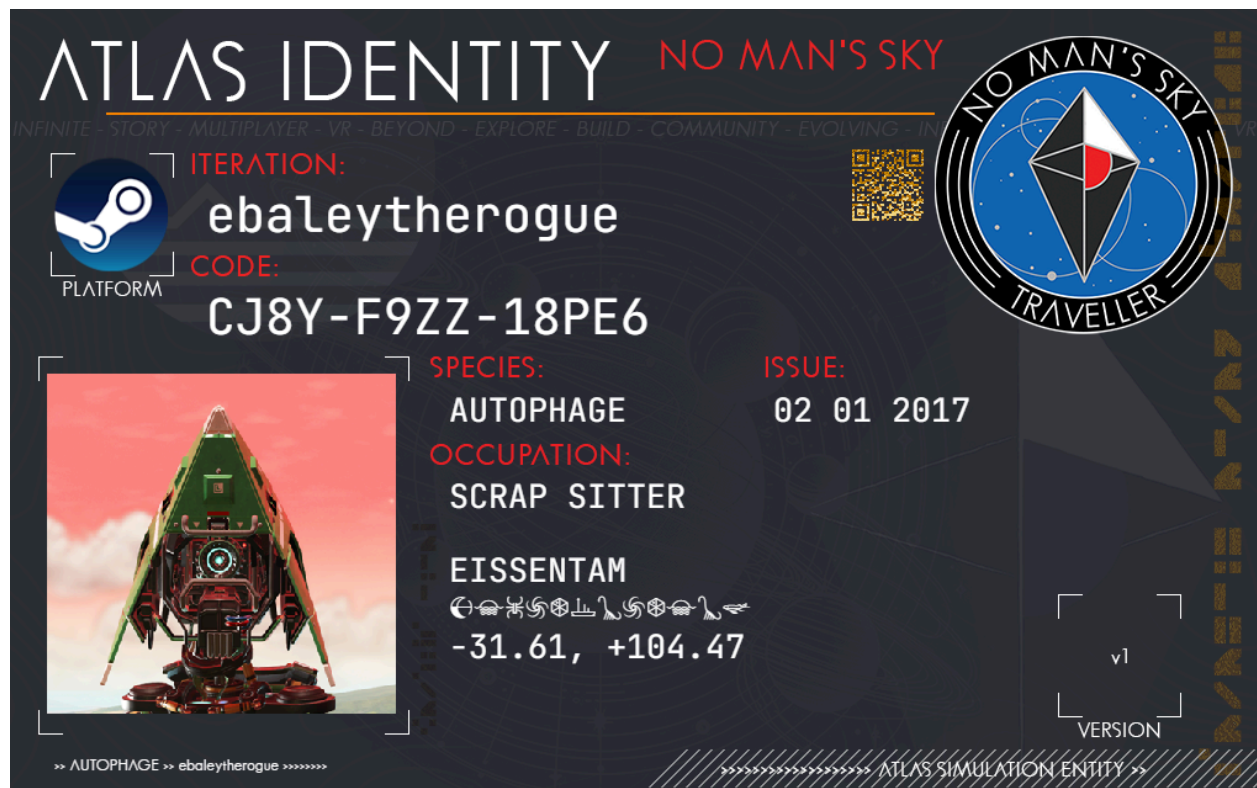
Name	Date modified	Type	Size
 nms_bases_duplicate_names.csv	3/25/2026 10:02 AM	CSV File	2 KB
 nms_bases_grouped.csv	3/25/2026 10:02 AM	CSV File	46 KB
 nms_bases_master.csv	3/25/2026 10:02 AM	CSV File	49 KB
 nms_bases_summary.txt	3/25/2026 10:02 AM	Notepad++ Docu...	3 KB

You can also run this manually if you'd prefer:

In Windows Explorer, type POWERSHELL in the address bar of the folder and press <ENTER>. A PowerShell window should open into that folder.

Press <ENTER> to generate the files.

Dan Reams a.k.a. ebaleytherogue in No Man's Sky.



Tips

Base-Names in Reports Might be Out-of-Date if Not Uploaded

I saw (2026-04-02) that one of the bases in my reports files had a name older than the name as it appears in the online game. This was opposite to what I had imagined, as I'd assumed the local file would always have the latest and greatest values, with the online version only being updated either by manual base-upload or via some sort of timed synchronization.

I journeyed to the base, initiated an 'Upload Base', and my reports were then in sync with the online game.

Because I had renamed the base more than five days and 10-12 game sessions prior, and that sync had not already occurred, I think it safe to say that such synchronization only occurs via a manual base upload.

One might surmise that the online game is housing information, newer than the local gamesave, but that is a premature conclusion. Instead, having pored over the JSON file, I think the real issue is the changed name is present in the game-save, in some other location than the Teleporter list, and that the teleporter list -- which I use as the 'base name' -- is the one that gets updated after the manual upload. Bases are fickle things in the files, which is why I use the Teleporter list both for basenames and for "base" coordinates.

TL;DR - Simply make sure to upload a base after you rename it, and you can avoid this. I thought I always did, but I'd clearly missed at least one.

Change Log

I opt to include the change log items, in this PDF, to keep me honest. It forces me to update the documentation to reflect the changes I make to the script and its related files.

Apr 8, 2026

Duplicate Bases Falsely Inflating the Base Count

In `nms_bases_grouped` I noticed, whilst scrolling, that some bases had the same Computer Coordinates. It looked like old base names were mixed in with later iterations of the same base.

I created a pivot table, to group bases by Computer Coordinates, and discovered many duplicates. My total number of bases is thus falsely inflated.

I am currently going through the code to change to using a persistent base identify as the source of truth and then supplementing the winning records with information from the teleporter list. I think I can do that because the real base record lives in PersistentPlayerBases, which has the authoritative Name, LastUpdateTimestamp, and the Objects list containing the base computer ^BUILDSAVE.

This will take a while. WORKING ON THIS. I will publish this document to the folder but not bother including it in the ZIP file until I have a fix in place.

Mar 25, 2026

On-Planet Locations Added

I added two columns.

- Computer Coordinates - This was my attempt to provide 'Current Location' information to be used in one's scanner, or starship, or exocraft, or anywhere that has such functionality, to precisely locate a base. However, the information within the .JSON is not consistent for all bases, for reasons I won't get into here, so I added...
- Teleporter Location - This information IS consistent for every teleporter. So, unless you don't have a teleporter at your base, this should be all you need. However, I am leaving the Computer Coordinates value there a) as an interesting item, and b) for those edge cases where the value is both populated and you do not have a teleporter at that base.

Example Below:

G	H
Computer Coordinates	Teleporter Coordinates
-57.33, -79.92	-57.41, -80.06
	-55.22, -159.99
-57.33, -79.92	-24.15, -34.75
-27.88, +22.71	-27.88, +22.70
+49.95, +106.94	+49.96, +106.94
	-11.00, +45.96
	-15.71, -94.04
	+14.10, -156.00
	-45.52, -30.48
	-73.85, -30.36
+32.36, +127.63	+32.36, +127.65
	+57.72, -90.01

Teleporters Always Have Coordinates in the Teleporter System, so it's a good value to use.

Changed Reports Folder to always be "nms_base_reports"

During my testing I had included the script's version in the folder name so I would not get confused as to which version of reports I was reading. Now that the system is stable, changing the folder to a static value makes more sense.

Lines changed to:

```
default="nms_base_reports",  
help="Output directory (default: nms_base_reports)",
```

I changed all references to the reports folder, in the instructions portion of this document, to reflect this new name.

Changed Main Script Name to NOT Include Version Number and .BAT file to Reference This New Name

I did this so you would not have to rename or edit these files each time I release a new script version.

I am including both a versioned-named file and a non-version named file in the ZIP file, for your convenience.

Name

 extract_nms_bases.py

 extract_nms_bases_v8.py

In this example, the files are identical.

The instructional portions of this document have been edited to reflect this.

Changed the Called JSON Filename to save.hg.json from save8.hg.json

I did this so that the .bat file does not need to be changed each time your JSON filename changes. Instead, you will need to change the JSON filename to this new value prior to running

the script. Reasoning: It's easier and less risky to change a JSON filename than open a batch file to change the called filename in there.

The instructional portions of this document have been edited to reflect this.